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Class Work Linux Lab: Files, APT, and C Programming

I. FILE OPERATIONS

mkdir ~/linux_lab

-created a new directory named linux_lab in the home folder. We used this directory to keep all the files and programs created during the lab in one place.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```

cd ~/linux_lab

-moved us from the home directory into the newly created linux_lab directory. After this command, the files we created were stored inside the lab folder.

```
aayushdas@Ubuntu:~$ mkdir ~/linux lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```

nano greeting.txt

-opened the Nano text editor and created a new file called greeting.txt. We then entered "Hello Linux!" into the file and saved it.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```



The screenshot shows the nano text editor interface. At the top, it says "GNU nano 8.7.1". Below that, the text "Hello Linux!" is entered and followed by a cursor. The editor's status bar at the bottom shows "aayushdas@Ubuntu:~/linux_lab\$".

cat greeting.txt

-displayed the content of the greeting.txt file directly in the terminal. This allowed us to check that "Hello Linux!" had been saved correctly.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```

cp greeting.txt backup.txt

-made a copy of the original greeting.txt file. The new copy was named backup.txt, while the original file remained unchanged.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```

mv backup.txt old_greeting.txt

-changed the name of the copied file from backup.txt to old_greeting.txt. The file itself was not deleted; only its name was changed.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```


rm old_greeting.txt

-removed the old_greeting.txt file from the directory. After running this command, the file was no longer available in the lab folder.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```

mkdir temp_dir

-created a new directory named temp_dir. This was only used as a temporary directory for practicing directory removal.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdor temp_dir
```

rmdir temp_dir

-removed the temp_dir directory because it was empty. This command is useful when we want to remove an empty directory without deleting any files inside it.

```
aayushdas@Ubuntu:~$ mkdir ~/linux_lab
aayushdas@Ubuntu:~$ cd ~/linux_lab
aayushdas@Ubuntu:~/linux_lab$ nano greeting.txt
aayushdas@Ubuntu:~/linux_lab$ cat greeting.txt
"Hello Linux!"
aayushdas@Ubuntu:~/linux_lab$ cp greeting.txt backup.txt
aayushdas@Ubuntu:~/linux_lab$ mv backup.txt old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ rm old_greeting.txt
aayushdas@Ubuntu:~/linux_lab$ mkdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ rmdir temp_dir
```

ls -l greeting.txt

-displayed detailed information about the greeting.txt file. The output showed details such as the file permissions, owner, size, and modification time.

```
aayushdas@Ubuntu:~/linux_lab$ rmdir temp_dir
aayushdas@Ubuntu:~/linux_lab$ ls -l greeting.txt
-rw-r--r-- 1 aayushdas aayushdas 14 Oct 10 10:10 greeting.txt
```

II. APT PACKAGE MANAGEMENT

sudo apt update

-updated the list of available software packages on the Linux system. We did this before installing the required tools so that Linux could use the latest available package information.

```
aayushdas@Ubuntu:~/linux_lab$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:5 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.4 kB]
Get:7 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [601 kB]
Get:8 http://np.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.3 kB]
Get:9 http://np.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [409 kB]
Get:10 http://np.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [79.5 kB]
Get:11 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [268 kB]
Get:12 http://np.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:13 http://np.archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.6 kB]
Get:14 http://np.archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3,036 B]
Get:15 http://np.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,056 B]
Fetched 2,161 kB in 3s (692 kB/s)
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install build-essential

-installed the basic development tools needed for the lab. This package includes GCC, which we needed later to compile our C program.

```
aayushdas@Ubuntu:~/linux_lab$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30
  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-modules-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 9
aayushdas@Ubuntu:~/linux_lab$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```


gcc --version

-checked whether GCC was installed correctly on the system. It also displayed the version of the GCC compiler currently installed.

```
aayushdas@Ubuntu:~/linux_lab$ sudo apt install build-essential
build-essential is already the newest version (12.12ubuntu2.26.04.2).
The following packages were automatically installed and are no longer required:
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30
  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-modules-7.0.0-30-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 9
aayushdas@Ubuntu:~/linux_lab$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

apt search python3

searched the available Linux packages related to Python 3. This was an example of how we can search for software packages using APT.

```
aayushdas@Ubuntu:~/linux_lab$ apt search python3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
  Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
  Python 3 packages for astronomy

bamkit/resolute 0.0.1+git20170413.ccd079d-3build1 all
  tools for common BAM file manipulations

beancount/resolute 3.2.0-2build1 all
  Double-entry accounting from text files

berrynet-dashboard/resolute 3.10.2-2build1 all
  deep learning gateway - python3 modules

brlTTY/resolute,now 6.7-1ubuntu6 amd64 [installed,automatic]
  Access software for a blind person using a braille display

budgie-app-launcher-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
  Applet to provide an alternative means to launch applications

budgie-brightness-controller-applet/resolute-updates 2.2.3-1ubuntu1.1 amd64
  Applet to control the brightness of the screen
```


III. C COMPILATION

A. Step 1: Write the Code

```
#include <stdio.h>

int main() {
    int a = 10, b = 5;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

B. Step 2: Compile and Run

nano hello.c

-opened the Nano editor and created a new C source file named hello.c. We then wrote the simple C program inside this file and saved it.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ls -l hello_program
-rwxr-xr-x 1 aayushdas aayushdas 16384 Nov 14 10:10 hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

gcc hello.c -o hello_program

-compiled the C source code written in hello.c. If there were no errors, it created a new executable file called hello_program.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ls -l hello_program
-rwxr-xr-x 1 aayushdas aayushdas 16384 Nov 14 10:10 hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

ls -l hello_program

-checked whether the compiled executable had been created successfully. The command also displayed detailed information about the new program file.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ls -l hello_program
-rwxr-xr-x 1 aayushdas aayushdas 16384 Jan 10 10:10 hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

./hello_program

-ran the C program that we had just compiled. The program displayed "Hello, Linux Lab!" and calculated the sum of the two numbers given in the code.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ls -l hello_program
-rwxr-xr-x 1 aayushdas aayushdas 16384 Jan 10 10:10 hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

gcc -Wall hello.c -o hello_program_warn

-compiled the C program again, but this time with warnings enabled. The -Wall option helps us notice possible problems or mistakes in our code while compiling.

```
aayushdas@Ubuntu:~/linux_lab$ gcc -Wall hello.c -o hello_program_warn
aayushdas@Ubuntu:~/linux_lab$ ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

./hello_program_warn

-ran the version of the program that was compiled with warnings enabled. Since there were no problems with our code, the program ran normally and displayed the expected output.

```
aayushdas@Ubuntu:~/linux_lab$ gcc -Wall hello.c -o hello_program_warn
aayushdas@Ubuntu:~/linux_lab$ ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15
```

C. Step 3: Modify and Recompile

nano hello.c

-opened the C source file again so that we could make changes to the program. We changed the value of a from 10 to 200 and then saved the file.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc -Wall hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300
#include <stdio.h>

int main() {
    int a = 100, b =200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

We used **gcc -Wall hello.c -o hello_program** to compile the modified C program again. Recompiling was necessary so that the changes we made to the source code were included in the executable program.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc -Wall hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300
```


./hello_program

-ran the updated program after recompilation. The output now showed the new sum using 200 instead of the previous value of 10, demonstrating that our changes had been successfully applied.

```
aayushdas@Ubuntu:~/linux_lab$ nano hello.c
aayushdas@Ubuntu:~/linux_lab$ gcc -Wall hello.c -o hello_program
aayushdas@Ubuntu:~/linux_lab$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 is 300
```

IV. CLEANUP AND VERIFICATION

ls -lh ~/linux_lab/

-displayed the files inside our lab directory along with their sizes and other details. This allowed us to check what files and compiled programs had been created during the lab.

```
aayushdas@Ubuntu:~$ ls -lh ~/linux_lab/
total 40K
-rw-rw-r-- 1 aayushdas aayushdas 15 Sep  8 05:51 greeting.txt
-rw-rw-r-- 1 aayushdas aayushdas 164 Sep  8 06:16 hello.c
-rwxrwxr-x 1 aayushdas aayushdas 16K Sep  8 06:18 hello_program
-rwxrwxr-x 1 aayushdas aayushdas 16K Sep  8 06:12 hello_program_warn
```

cat ~/linux_lab/hello.c

-displayed the final C source code directly in the terminal. We used this as a final check to make sure the code was saved correctly.

```
aayushdas@Ubuntu:~$ cat ~/linux_lab/hello.c
#include <stdio.h>

int main() {
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
```

rm -rf ~/linux_lab/

-removes the entire linux_lab directory along with everything inside it.

```
}
aayushdas@Ubuntu:~$ rm -rf ~/linux_lab
aayushdas@Ubuntu:~$
```